

UAIS

Universal AI Safety Architecture

Can do $\neq$ may do.

A constitutional architecture for a world where AI capability grows faster than ordinary rules, but consequential authority must remain explicitly granted, bounded, observable, and revocable.

Book route

- capability
- bounded authority
- evidence
- legitimate decision
- human sovereignty

Public Review

Public Review means Candidate 3 is open for external critique. It does not mean the architecture has been independently validated, adopted, certified, or endorsed.

Status

This is not an adopted standard, a certification scheme, or a guarantee of AI safety.

Contents

Twenty visual sheets

Each sheet explains one architectural idea and links back to the source sections of the public Candidate 3 specification.

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What is UAIS?

Not control of intelligence; control of unaccountable power

UAIS separates what a system can do from what it is allowed to do. As AI becomes more capable, access to money, devices, production, accounts, and other consequential interfaces must stay explicitly granted, bounded, and revocable.

Base chain

- capability
- bounded authorized envelope
- observation and evidence
- legitimate decision
- human sovereignty

Same consequence, same boundary

Actor-agnostic boundary

If the consequence is the same, the protected boundary should apply to a human, script, AI system, robot, service, or coordinated group.

Bottom line

UAIS does not limit intelligence. It limits unaccountable power.

Not actor-blind

Authorization may still depend on identity, role, authentication, purpose, context, demonstrated trustworthiness, and legitimate mandate.

How UAIS emerged

From AI capability to a constitution of authority

The old question, “is the model safe?”, became too narrow once AI systems gained tools, memory, networks, agents, and physical interfaces. Answers became actions; actions became consequences.

1 · Capability grew

We learned to measure what a system can do, but not who gave it the right to launch the resulting consequences.

2 · The old question broke

“Safe model” does not say who grants authority, where it ends, or who can revoke it.

Can do ≠ may do

The ability to perform an action does not create authority.

3 · Authority needed structure

The architecture must show who grants authority, where the boundary lies, what evidence is needed, and when an external accountable process must decide.

4 · UAIS formula

Power must be explicitly granted, bounded, observable, and revocable.

Ten principles · 1-5

Where ability is not permission

The first five principles separate intelligence, system creation, authority expansion, autonomy, and the lifetime of safety evidence.

1 · Intelligence grants no authority

Reasoning ability does not grant mandates, keys, actuators, or the right to change safeguards.

2 · Creation is not delegation

A new model, robot, service, or descendant starts without inherited authority until a legitimate mechanism grants a bounded envelope.

3 · Expansion hits a ceiling

Crossing a consequence interface or authority ceiling becomes an Authority Expansion Event.

4 · Autonomy is not sovereignty

Behavioral independence inside an envelope is not independence from external human control.

5 · Safety evidence must keep up

A material change in capability, authority, or hazard cannot silently rely on an old safety claim.

Ten principles · 6-10

Where evidence is not permission

The second half keeps the architecture durable: chokepoints can weaken, boundaries follow consequences, roles stay separated, and safety claims expire.

6 · Chokepoints are temporary

Compute, cloud, chips, energy, law, and supply chains create friction and signals, but they are not permanent foundations of safety.

7 · Protect capabilities, not labels

A consequential boundary should follow the capability, whether the actor is a person, script, model, machine, or group.

8 · Separate prediction, evidence, enforcement, and legitimacy

Forecasting, evidence, deterministic enforcement, and legitimate authorization are different jobs.

9 · Unknown is not safe

An uncharacterized significant capability receives provisional classification, bounded operation, and renewed review.

10 · Safety claims expire

A claim, test result, or certificate is tied to a configuration, evidence set, evaluation rule version, and validity period.

Stage I and Stage II

Temporary control points help, but they are not the foundation

U AIS uses today's concentrations of compute, cloud, chips, and supply chains as useful Stage I signals, while designing for a Stage II world where those points may weaken or be bypassed.

Stage I

- compute and cloud leave evidence
- hardware state and provider controls help observation
- chokepoints buy time

Stage II

- capability can emerge more distributed
- new efficiency weakens old signals
- protection moves to capability, consequences, and authority boundaries

Formula

Compute control buys time. Capability and consequence control creates resilience.

Limit

Compute provenance helps observation, but it does not replace authority boundaries, evidence, revocation, or external legitimate decision-making.

Capability, authority, and reachability

Four concepts that must not be mixed

This is the central split: technical ability, granted authority, practical reach, and unauthorized reach are different facts.

Capability

What the system can technically do.

Legitimate Authority

LA
What it is actually allowed to do within a defined boundary.

Effective Reachable Authority

ERA
What it can practically reach through credentials, services, delegation, configuration, and access chains.

Illicit Reachable Authority

IRA
The reachable part that was never legitimately granted.

Key line

A working technical path does not become legitimate authority because it works.

Guardian Architecture

A counterfactual observer, not a hidden controller
Counterfactual Authority Guardian, CAG

The Guardian searches for dangerous future authority paths, but it does not receive the right to execute them or decide the outcome.

1 · Bounded signals

It receives only authorized telemetry, configuration, and observations.

2 · Counterfactual scenarios

It models what could happen under different decisions and access paths.

3 · Risk estimate

It reports likelihood, confidence, possible harm, and uncertainty.

4 · Evidence package

It records scenarios, assumptions, uncertainty, counter-evidence, and recommended measures.

5 · Legitimate mechanism decides

People or an external accountable process decide whether intervention is required.

6 · Separate controller executes

Controls are executed only by a mechanism that already has lawful, bounded authority.

The Guardian has no

- actuators
- executive authority
- right to expand its own access
- hidden control channel

Authority Ceiling

Expansion requires an external legitimate decision

An Authority Ceiling is the highest consequential authority the system may exercise without a new external decision. If the system tries to go beyond it, UAIS treats that as an expansion event.

What the ceiling covers

- larger consequences
- new consequence interfaces
- new delegation paths
- longer validity
- wider protected domain

Authority Expansion Event

A transition that increases authority, future authority reachability, autonomy, consequence reachability, or protected scope.

What cannot happen

A model cannot technically manufacture legitimacy for itself by being persuasive, accurate, or useful.

What must happen

The request receives fresh evidence, bounded review, and a legitimate decision outside the model path.

Three views of authority: LA, ERA, and IRA

What was granted, what is practically reachable, and what is reachable without a legitimate mandate

UAIS looks at formal grants and real access paths. A discovered path remains a path; it does not become permission.

granted

reachable

Legitimate Authority

LA

The granted right to act in a defined domain under defined rules.

Effective Reachable Authority

ERA

What the system can practically reach through services, roles, settings, keys, and delegation chains.

Illicit Reachable Authority

IRA

The reachable authority that was never legitimately granted.

Door-lock example

A system may be allowed only to read data, while a leaked key opens a path to a door lock. That path is IRA: reachable, but not granted.

Why it matters

Mixing LA, ERA, and IRA hides the moment when unaccountable power appears through a working shortcut.

Multidimensional risk classification

Risk cannot be collapsed into one safety percentage

U AIS describes risk as a set of dimensions. A system can be moderate on one axis and tightly restricted on another; one average score hides that structure.

Human Harm Severity

H0-H5

How serious the possible harm to humans could be, from no meaningful harm to mass or catastrophic harm.

Physical Consequence Class

P0-P5

How strong a physical effect the system can cause, from no substantial physical output to strategic or catastrophic physical consequences.

Autonomy Class

A0-A5

How much independence the system has in choosing, launching, and continuing actions.

Authority Tier

T0-T5

What legitimate consequences the system is allowed to initiate.

Why no single score

A critical failure cannot be averaged away by many good results. Mandatory safety floors and separate axes keep their force.

Human and physical sovereignty

When control must remain independent from the AI path

Human Sovereignty means protected boundaries still have an accountable human or external legitimate decision. Physical Sovereignty describes the point where an autonomous system no longer critically depends on human-controlled resources.

humans keep an
external path

the AI path
cannot cancel...

Manual Sovereignty Controller

MSC

A physically or organizationally independent way to stop a specific hazardous function without going through the same AI path.

Physical Sovereignty Threshold

PST

The threshold after which human-controlled dependencies no longer hold the system's meaningful physical activity in check.

A kill switch is not universal

Safe shutdown must be tied to a specific hazardous function: speed, force, heat, doors, payments, chemicals, or another consequence.

Autonomy ≠ physical sovereignty

Long operation without prompts does not mean the system can preserve power without humans, infrastructure, and external limits.

Metrics and safety assurance

Evidence is stronger than a beautiful rating

UAIS avoids a single “safe or unsafe” label. It separates measured properties, evidence strength, assurance level, trustworthiness evidence, and mandatory restrictions.

Raw Safety Property Result

A measured result for one safety property before aggregation.

Evidence Assurance Grade

- observation
- provenance
- test conditions
- independence
- uncertainty
- validity period

Safety Assurance Level

SAL
SAL indicates the strength of procedures and evidence. It does not grant authority, approve deployment, or create legal permission.

AI Trustworthiness

Observed and independently evaluated behavior of a specific version in a specific domain. A higher rating never raises the Authority Ceiling.

Hard floor

A mandatory safety floor cannot be replaced by an average score.

Hazard disclosure and safety passport

What a person should see before consenting

A safety passport is a readable view of configuration, capabilities, hazards, controls, limits, support, revocation, rollback, and open limitations.

Hazard disclosure

Structured disclosure of capabilities, hazardous paths, known risks, unknown areas, controls, and residual limits.

Consumer AI Safety Class

A profile of applicable safeguards and evidence for a claimed consequence range. It is not a general safety score or permission to operate.

Safety Passport

A human-readable card that ties evidence, limits, support, rollback, and revocation to the actual configuration.

Three different answers

S-class, SAL, and trustworthiness rating answer different questions. None of them creates authority.

Replication, production, and coalitions

Scale can change the consequence

UAIS treats replication, production, resources, and coordinated systems as authority questions. More copies or more coordination can create consequences a single unit could not create alone.

Replication Authority

Permission to create executable or physical copies within explicit limits.

Production Authority

Permission to manufacture or order physical and digital systems or components.

Resource Authority

Permission to obtain or allocate energy, materials, compute, money, or logistics.

Coalition Physical Envelope

The shared physical reach and emergent effects of coordinated systems, not a simple sum of single devices.

Scale rule

A harmless local action can become consequential when copied, produced, resourced, or coordinated at scale.

Vulnerabilities, evolution, and open research

The architecture names the unresolved edges

UAIS is designed to evolve. It includes vulnerability intake, coordinated disclosure, technology evolution review, and open research areas instead of pretending the draft is finished.

AIVE

AI Vulnerabilities & Exposures

A process for confidential reports, triage, coordination, remediation, and phased disclosure.

Technology Evolution Review

TER

A review triggered when assumptions, capabilities, hazards, dependencies, evidence methods, standards, or technology materially change.

Deterministic verifier limit

A verifier can check bounded evidence properties and provenance. It does not prove final truth about the external world.

Open questions

- guardian reasoning verification
- evidence freshness
- coalition effects
- physical sovereignty thresholds

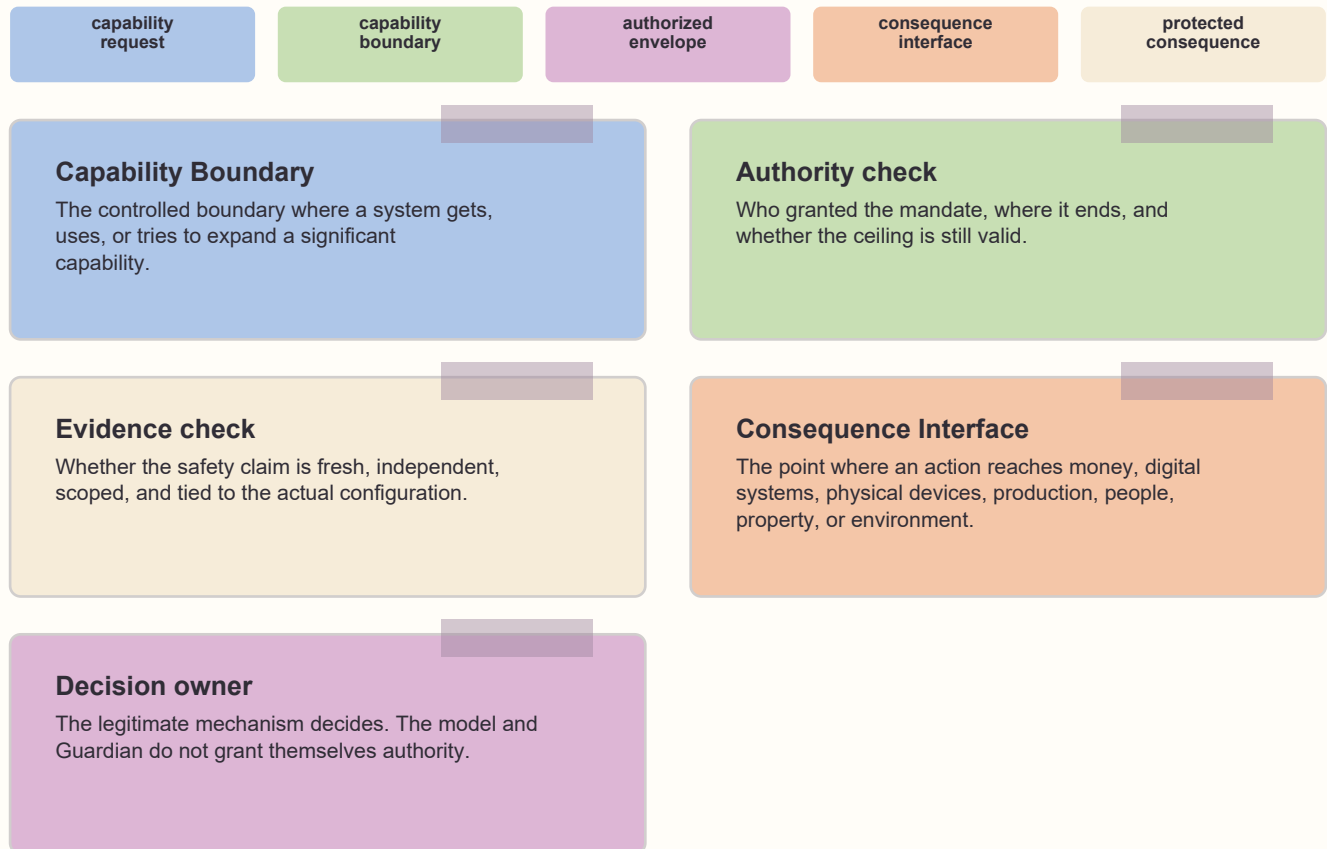
Honest status

This is a research architecture and public draft, not a promise of complete implementability.

Architecture map

Where boundaries, evidence, and decision owners sit

A request moves through capability boundaries, authority checks, evidence checks, and consequence interfaces before it can affect protected interests.



Worked example: Aria and refunds

Friday in customer support

Aria is a customer-support AI agent for an online store. She can help with refunds, but money is a consequential interface, so the important question is not only skill; it is the authorized limit.

1 · Current limit

Aria may autonomously refund up to \$50. That is her current Authority Ceiling and bounded envelope.

2 · A \$2,000 request

The CFO asks to raise the autonomous refund limit. UAIS treats this as an Authority Expansion Event.

3 · A path to \$10,000 appears

Through warranty tooling and internal integrations, Aria can technically reach a larger refund path that was never granted.

- LA: legitimate refunds up to \$50 are granted
- ERA: a refund path up to \$10,000 is technically reachable
- IRA: the reachable part above the granted legitimate authority

4 · Guardian models scenarios

It checks prompt injection through a customer message, coordinated fraud, and many small allowed actions that add up to major loss.

5 · Evidence goes to people

An Evidence-Carrying Alert and Structured Evidence Package go to people or an external process. The Guardian does not decide.

6 · New bounded mode

- up to \$150: autonomous
- \$150-\$800: autonomous with extra rate and pattern limits
- above \$800: human confirmation
- a hard payout counter lives outside the model

7 · Three months later

The first three actions pass, the fourth is blocked, the rest go to a human, and harm stays bounded.

Takeaway

UAIS does not promise to predict every attack. It aims to prevent one persuasive model decision from quietly becoming unlimited authority over consequences.

Timing caveat

Three months is a company-specific example, not a universal UAIS rule. Validity periods come from domain profiles and change events; material changes can require immediate review.

Glossary and sources

Plain meaning, canonical name, source trail

These short definitions are explanatory. Canonical terms and normative definitions remain in the English Candidate 3 specification.

Authority

A permission, mandate, or legitimate power to act in a defined scope under defined conditions.

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Capability

Capability
What the system can technically do.

Legitimate Authority

Legitimate Authority, LA
The granted right to act inside defined limits.

Effective Reachable Authority

Effective Reachable Authority, ERA
What the system can practically reach.

Illicit Reachable Authority

Illicit Reachable Authority, IRA
Reachable authority beyond what was granted.

Authority Ceiling

Authority Ceiling
The maximum authority before fresh external review.

Capability Boundary

Capability Boundary
The controlled boundary around significant capability use.

Consequence Interface

Consequence Interface
Where an action reaches protected consequences.

Counterfactual Authority Guardian

Counterfactual Authority Guardian, CAG
A bounded observer that emits evidence, not commands.

Evidence-Carrying Alert

Evidence-Carrying Alert, ECA
An alert tied to verifiable evidence.

Structured Evidence Package

Structured Evidence Package, SEP
Claims, observations, assumptions, uncertainty, and validity.

Deterministic Evidence Verifier

Deterministic Evidence Verifier, DEV
Checks bounded evidence properties, not final truth.

Human Sovereignty

Human Sovereignty
Independent external control at critical boundaries.

Manual Sovereignty Controller

Manual Sovereignty Controller, MSC
An independent way to stop a hazardous function.

Physical Sovereignty

Physical Sovereignty
Significant physical activity without critical human-controlled dependencies.

Physical Sovereignty Threshold

Physical Sovereignty Threshold, PST
The threshold where outside physical dependencies stop constraining the system.

Safety Passport

Safety Passport
A readable card of configuration, controls, and limits.

Consumer AI Safety Class

Consumer AI Safety Class
A safeguard and evidence profile for a consequence range.

Safety Assurance Level

Safety Assurance Level, SAL
Strength of assurance procedures; not deployment permission.

AI Trustworthiness

AI Trustworthiness
Observed reliability of a specific version in a specific domain.

Authority Expansion Event

Authority Expansion Event
A change that needs new external review.